

Building Bigger Shapes from Smaller Ones

Answer Key

Kindergarten–Grade 1

Quick Answers

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|--|--|
| 1. how many small squares = 2 | 6. how many more squares = 2 |
| 2. how many small squares = 3 | 7. how many small squares = 5 |
| 3. how many small squares = 6 | 8. (b) how many small squares it covers = 6, — |
| 4. (b) how many small squares it covers = 4, — | 9. (a) the symbol = =, — |
| 5. the symbol = > | 10. (a) how many more squares = 3, — |
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What is verified

6 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 4, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Kindergarten–Grade 1

1.G – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 14 tagged checks.

1. How many small squares cover the shape (two squares side by side)?

Solution: The thick outline holds one row with two small squares in it. Touch each once: one, two. 2

2. How many small squares cover the shape (the corner piece)?

Solution: Count the bottom row first — two squares — then the one sitting above the left-hand square. Two and one more. 3

3. How many small squares cover the rectangle?

Solution: This one has rows of equal length, so count one row and then the other rather than hunting around the figure: three in the bottom row and three in the top row. 6

4. Two matching triangles are joined along their longest edges. (a) Name the new shape. (b) How many small squares does it cover?

Solution (a) — instructor-judged. The two triangles fit together into a *square*: all four sides are the same length and every corner is square. Full credit for the word *square*, spelled any way a reader can sound out. *Rectangle* is true but less exact and earns half credit

— worth saying aloud that every square is also a rectangle. *Triangle* names the pieces, not the shape they build, and earns no credit.

Solution (b): Ignore the join line and count the small squares under the whole outline: two in the bottom row and two in the top. 4

5. Two shapes, A and B, were built from small squares. Which uses more? Write $<$, $>$, or $=$.

Solution: Shape A covers five small squares — a two-by-two block with one more on top of the left column. Shape B is a two-by-two block, so it covers four. Five is more than four, so the wide end faces A. 5 > 4

6. The dashed line shows the rectangle to be built; the shaded squares are already placed. How many more are needed?

Solution: Do not add them one at a time. The finished rectangle holds six small squares, three in each row, and four are already shaded, so what is missing is $6 - 4$. Checking by eye: the two unshaded cells sit in the top row. 2

7. How many small squares cover the plus-shaped figure?

Solution: Count the middle row across — three squares — then the one above it and the one below it. 5

8. Two matching corner pieces are fitted together. (a) Name the new shape. (b) How many small squares does it cover?

Solution (a) — instructor-judged. The two corner pieces interlock into a *rectangle*: two rows of the same length, with square corners, longer than it is tall. Full credit for the word *rectangle*, spelled any way a reader can sound out. *Square* earns no credit here, because the figure is longer than it is tall, and naming the corner pieces instead of the built shape earns no credit either.

Solution (b): The join line inside does not change the count. Three small squares in the bottom row and three in the top. 6

9. Two shapes, A and B, look very different. (a) Compare the number of small squares in each. (b) Explain how they can use the same number.

Solution (a): Shape A is two rows of four, so eight small squares. Shape B is a row of five with three above it, which is also eight. Neither is bigger. 8 = 8

Solution (b) — instructor-judged. A correct response says the squares were rearranged, not added to or taken from. Full credit for that idea in any wording — the same eight squares pushed into a different arrangement give a different outline while the amount stays put. Accept a count of both figures offered as the reason. Saying only that both counts came out the same, with no account of why that is possible, earns half credit. Claiming that the longer figure must be bigger earns no credit.

10. The dashed line shows the rectangle to be built; the shaded squares are already placed. (a) How many more are needed? (b) Explain how you worked it out.

Solution (a): The finished rectangle holds eight small squares, four in each row. Five are already shaded, so the number still missing is $8 - 5$. 3

Solution (b) — instructor-judged. A correct response gives a method rather than a count of one-by-one placements. Full credit for either of these in the child's own words: work out how many the finished outline would hold, then take away the ones already there; or count the empty cells inside the dashed outline directly. Accept a drawing of the missing

squares offered together with a count. A correct number reached only by guessing and checking earns half credit. Repeating the number from part (a) with no method earns no credit.